

Yuou Zhang

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My Website: ericzhang12138.github.io/Eric_website

EDUCATION

University of Edinburgh

Edinburgh, UK

MEng Computer Science and Electronics; GPA: 79/100

September 2022 – June 2027

- Key Modules: Software Design and Modelling, Algorithm and Data Structure, Object-Oriented Programming, Operating Systems, Computer Communication and Network, Distributed Systems, Computer Graphics, Databases, Computer Architecture, Machine Learning Practical, Machine Learning and Data Analysis
- Awards:
 - * Class Medal - Best Performing Student in E&CS (Top 1, 2022/23)
 - * Class Medal - Best Performing Student in E&CS (Top 1, 2023/24)

EXPERIENCE

Software Engineer Intern

October 2025 – Jan 2026

Nethermind

Edinburgh, UK

- Engineered a high-throughput biometric enrollment pipeline using a Producer-Consumer architecture with Python threading and asyncio. Decoupled hardware I/O from NPU inference to minimize latency, employing a leaky-bucket queue strategy that guarantees real-time processing of the freshest frames while maintaining a non-blocking event loop for the main application.
- Engineered TLS software layer with HSM private key operations.

AI Research Intern

June 2025 – Sep. 2025

University of Edinburgh

Edinburgh, UK

- Developed a multimodal machine learning system combining a pretrained image encoder (EfficientNet) and a pretrained text encoder (DistilBERT on OCR-extracted text) to improve fine-grained grocery product identification for visually impaired users
- Designed a feature fusion pipeline to integrate visual and textual cues from product packaging, aiming to resolve visual ambiguities and ultimately improving accuracy by 6.2%.

Firmware Engineer Intern

June 2024 – Sep. 2024

Metrol Technology

Aberdeen, UK

- Architected scalable MRAM validation firmware in C using design patterns (dependency injection, strategy pattern) to support 8+ testing algorithms across 10+ chip variants, achieving 98%+ fault coverage while reducing code duplication by 60%

PERSONAL PROJECTS

Distributed Filesystems [GitHub] | C++, gRPC, Docker, GitHub Action, Distributed System, Filesystem, Multithreading

- Built a fault-tolerant, AFS-inspired distributed file system in C++ with multi-server replication architecture and client-side whole-file caching, ensuring high availability and optimized read/write performance
- Implemented AFS-style callback mechanism for cache coherency with server-initiated invalidation, utilizing producer-consumer queue architecture for multi-threaded concurrent request processing
- Integrated with FUSE for seamless user experience, implemented features like atomic save on multiple platforms (Linux and MacOS) and deployed on Raspberry Pi using Docker for service orchestration, providing centralized file storage with cross-device access across home network clients, consistency guarantees, and automatic failover

Virtual Memory and Cache Simulator | C, Computer Architecture, Operating Systems, LRU/FIFO

- Built a complete virtual memory system in C with cache, TLB, main memory, page table and disk that simulates memory hierarchy hits/misses for 32-bit addresses
- Implemented LRU and FIFO replacement policies for cache and page management with comprehensive error handling for invalid memory accesses and page faults

TECHNICAL SKILLS

Languages: Modern C++, Python, Java, C, SQL, Bash, Haskell

Frameworks & Tools: Docker, Kubernetes, CI/CD, Ansible, PostgreSQL, Redis, Git/GitHub/Gitlab, GitHub Actions, CMake, CUDA, gRPC, FastAPI, PyTorch, NumPy, Pandas, Anaconda, FUSE, Azure

Systems & Platforms: Unix/Linux, Linux kernel